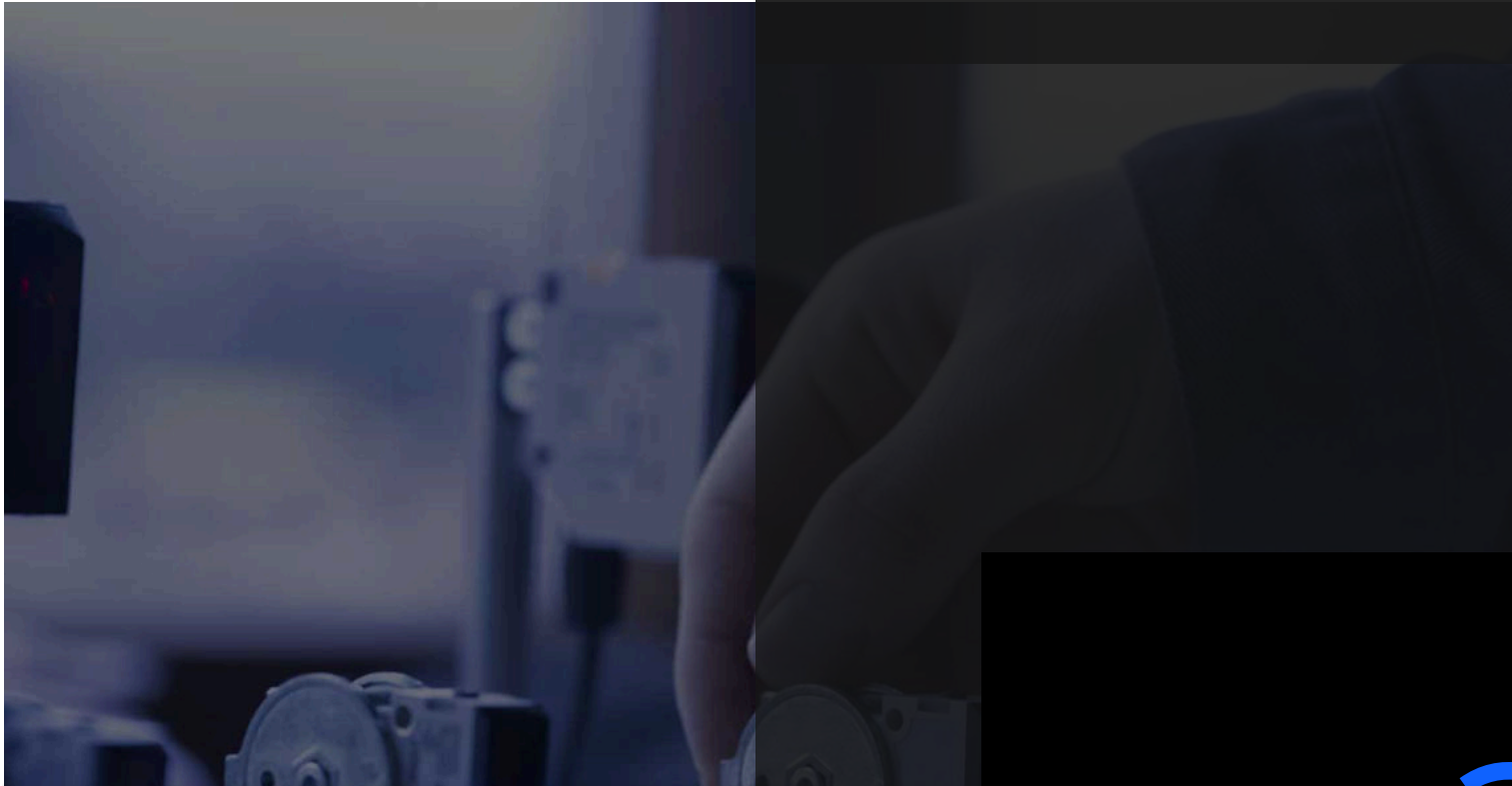


Using cognitive capabilities to spot the imperceptible

Novate Solutions gains 24x7 process reliability



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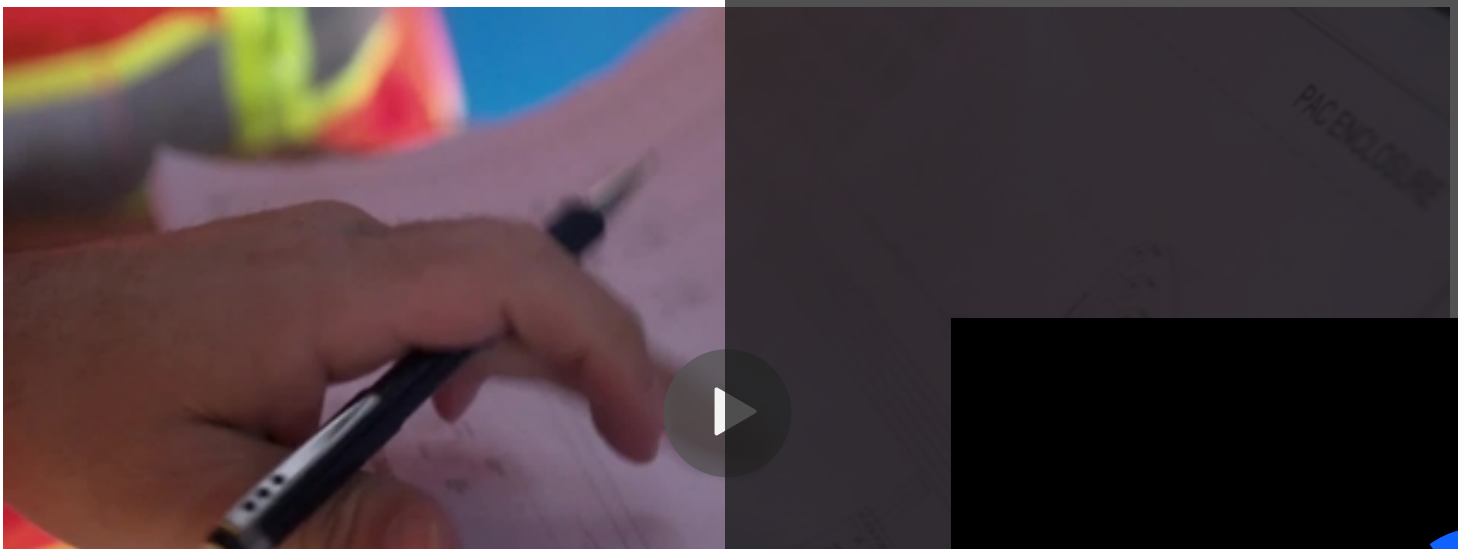
AI

Process failure costs are measured by the minute, so time is of the essence. Companies must be able to adjust to changing circumstances, and fast. What if you had a process control system that could spot issues before a catastrophic event occurred?

That's just what engineering and technology services firm Novate Solutions, Inc. sought. The company wanted to:

- Gain insights from machines that had been producing data for decades
- View how assets perform in real time
- Identify the root causes of failure much faster
- Reduce operating costs and downtime

Enterprise assets are equipped with sensors generating hundreds of thousands of data points. But Novate knew that data without context or correlation with other data can't provide meaningful information or process-wide insights to help ensure equipment is up and running. That's typically the job of the engineer in charge of evaluating the data.



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Real-time monitoring

IBM® Maximo® Monitor shows how assets perform in real time and enables 30% improvement in product quality

Waste reduction

Novate gets new data from old sensors and achieves 15% reduction in waste

“ Process control systems are designed to monitor and manage processes, but not to monitor and manage the underlying assets. But the data is there if you can capture it and you know what to do with it.”

Rob Mora

Executive Vice President of Business Development
Novate Solutions, Inc.

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maintenance.



Predictive insights: more than business as usual

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Although industrial manufacturers use sensor technology every day, they are also heavily tasked with day-to-day operations. To lighten their load, they need a scalable remote monitoring service that identifies system anomalies and predicts issues before they become a problem. To address this need, Novate is using AI and [IBM Maximo Monitor](#) software to develop a revolutionary new approach to improve process reliability and ultimately enhance industrial process control.

Novate is integrating its customers' control system infrastructures and process data with IBM cloud-based AI technology and Maximo Monitor, which is incorporated directly into its Support Operations Center. By doing so, the company can analyze the data in search of anomalies that would otherwise be imperceptible to a typical control system. If one is found, the system alerts engineers at Novate's Support Operations Center.

Then, drawing on their years of process experience, familiarity with customer operations and knowledge of engineering principles, Novate's engineers determine whether the anomaly indicates a problem. If necessary, they will then work with company technicians to prevent a predicted failure and avoid downtime.

"Ultimately, we want to train the AI to make the system smarter," explains Rob Mora, Executive Vice President of Business Development at Novate. "With help from IBM Watson-enabled AI, our service will teach systems to recognize events as they are developing."

But not all customers have access to the world-class AI at IBM or understand the best way to implement and extract value from it. In addition, it may not yet fit in a customer's technology roadmap. So to employ AI, they must first develop a solid use case.

Mora says that some of Novate's customers are extraordinarily sophisticated and some are just beginning their transformation journey. He visited a state-of-the-art, flagship

facility for a global food producer that makes around 400 tons of cookies a day. The facility produces 20 tons of cookies to unplanned downtime every day. The facility had purchased IBM technology solutions to improve efficiency and invested in 3D point cloud solutions to map some of the lines to reduce unplanned downtime. But it required a lot of time and effort to get the technology working. Mora remarks, "It creates a much broader field of view, taking in all the data. And it can be put into application without

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any capital investment or training through the use of our platform. Ultimately, AI allows the process systems to become self-aware and to ‘call out’ to an engineer and say, ‘You’re not looking over here, but something is not right and you ought to take a look.’ This takes plant reliability and support to another level.”

“ With help from IBM Watson®-enabled AI, our service will teach systems to recognize events as they are developing.”

Rob Mora
Executive Vice President of Business Development
Novate Solutions, Inc.

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Novate didn't create its solution simply to help manufacturers do what they do more efficiently but rather to enable true predictive capabilities. The solution can see issues that aren't normally monitored and alarmed.

Mora offers an example: “We were recently working with a food production facility on assets on the packaging line when the behavior of a servo motor in a piece of packaging equipment at the end of a production line came to our attention. What our system noticed was an increasing average torque value over time.”

Because the increasing torque value was unexpected behavior, the Novate solution flagged the equipment for inspection. Novate engineers contacted the food production ground crew. In the course of analyzing the packaging equipment, the crew determined that the oil had not been changed as the maintenance logs had suggested and had completely degraded, causing the motor to continuously work harder and harder.

“Our solution could see the client’s valuable information in the data even though the system and equipment were working properly,” says Mora. “There was no way the food production facility would have detected it. That’s something AI was able to notice and our engineers were able to decipher.”

As Novate continues to work with industrial manufacturers and employ AI-enabled industrial process control, the company will continue its partnership with IBM. “When we first connected with IBM, we were surprised by their willingness to take risks. They weren’t worried about immediate return on their investment. Their goal was innovation. Additionally, everyone at IBM has been collaborative. Our experience has been consistent with every contact,” says Mora.

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About Novate Solutions, Inc.

Founded in 2001 and based in West Sacramento and Livermore, California, [Novate](#) is an engineering and technology services firm that provides professional services to industrial manufacturers. Novate offers mechanical and electrical engineering, broad controls and automation integration expertise, turnkey system deployment capabilities and defense-in-depth industrial networking and cybersecurity solutions.

Solution component

IBM® Maximo® Monitor

→

Take the next step

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IBM Blog

What is predictive monitoring and how to make it happen

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